

EEE 103IL: Computer Programming Lab

Lab 05

Lab Tasks

1. Write a C program that reads 10 numbers from the user. The program then reads a number between 0 to 9, and shows the number at the corresponding index number. For instance, if the array is `a` and the user enters 3, your program should print the value `a[3]`.
2. Write a C program that reads 5 numbers from the user and prints the sum, followed by all numbers.
Input: 10, 20, 30, 40, 50
Output: Sum: 150
Numbers: 10, 20, 30, 40, 50
3. Write a C program that reads 5 numbers from the user, and then prints them in reverse order.
4. Write a C program that reads 10 numbers from the user, and then prints only the even numbers in reverse order.
5. Write a C program that reads 10 numbers from the user. Then read another number from the user, and print
"Yes found at index X" (here X is the index of the number in the array) if this number exists and print "No" otherwise.
6. Repeat task 1 but now put a validation for the last input. So after entering the 10 numbers, when the user is prompted to enter a number between 0–9, if the user enters any number less than 0 or greater than 9, keep asking to enter a valid number and show a message: "Enter a number between 0-9".
7. Write a program in C to concatenate one string with another string. For example: String 'a' contains "foot", string 'b' contains "ball", then string 'c' should contain "football".
8. A palindrome is a word, number, or sequence of characters which reads the same backward as forward, such as "madam". Write a C program to determine if a word is a palindrome or not.
9. Write a C program where the user inputs a sentence followed by a word. Your task is to count how many times the given word appears in the sentence.

Sample Input:

Sentence: This is a test. This test is simple.

Word: test

Sample Output:

Occurrences of "test": 2